Course overview

Introduction to computer programming

Management and Artificial Intelligence

LUISS

Greetings, folks!

Hello students!

Welcome to the *Introduction to Computer Programming* course!

Web Resources and More

Resources

Web Resources you should have access to

- LUISS Learn
- Course Syllabus
- Rooms and Timing

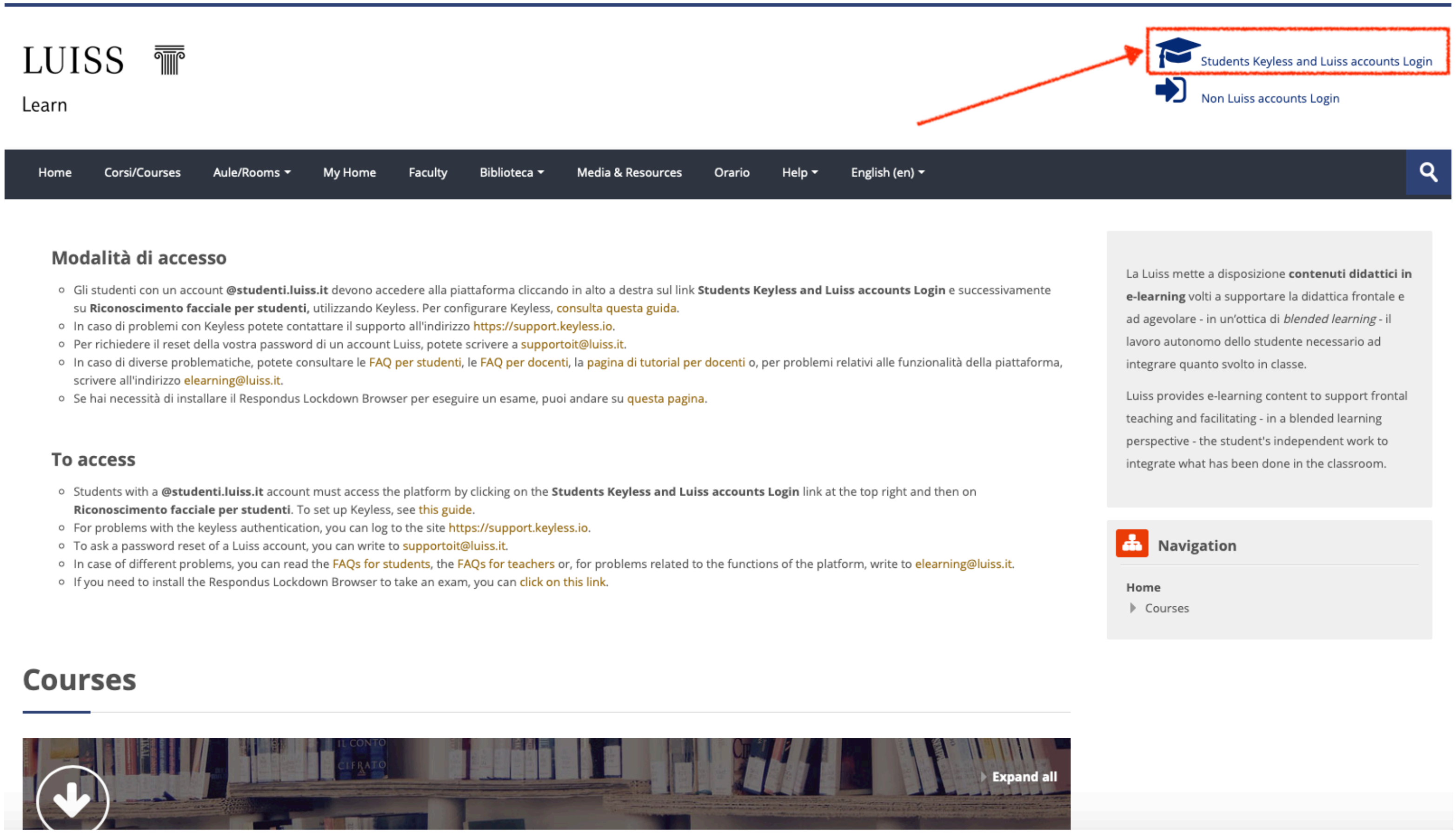
e-Learning platform

LUISS Learn is our e-Learning platform.

It is important that you get access!

You'll find there:

- Slides and other teaching material
- Exercises
- Details on exam dates, reservations, results (besides standard Luiss channels)
- Communications from teacher and teaching assistants



LUISS

Learn

Students Keyless and Luiss accounts Login

Non Luiss accounts Login

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Modalità di accesso

- Gli studenti con un account **@studenti.luiss.it** devono accedere alla piattaforma cliccando in alto a destra sul link **Students Keyless and Luiss accounts Login** e successivamente su **Riconoscimento facciale per studenti,** utilizzando Keyless. Per configurare Keyless, consulta questa guida.
- In caso di problemi con Keyless potete contattare il supporto all'indirizzo https://support.keyless.io.
- Per richiedere il reset della vostra password di un account Luiss, potete scrivere a supportoit@luiss.it.
- In caso di diverse problematiche, potete consultare le FAQ per studenti, le FAQ per docenti, la pagina di tutorial per docenti o, per problemi relativi alle funzionalità della piattaforma, scrivere all'indirizzo elearning@luiss.it.
- Se hai necessità di installare il Respondus Lockdown Browser per eseguire un esame, puoi andare su questa pagina.

To access

- Students with a **@studenti.luiss.it** account must access the platform by clicking on the **Students Keyless and Luiss accounts Login** link at the top right and then on **Riconoscimento facciale per studenti**. To set up Keyless, see this guide.
- For problems with the keyless authentication, you can log to the site https://support.keyless.io.
- To ask a password reset of a Luiss account, you can write to supportoit@luiss.it.
- In case of different problems, you can read the FAQs for students, the FAQs for teachers or, for problems related to the functions of the platform, write to elearning@luiss.it.
- If you need to install the Respondus Lockdown Browser to take an exam, you can click on this link.

La Luiss mette a disposizione **contenuti didattici in e-learning** volti a supportare la didattica frontale e ad agevolare - in un'ottica di *blended learning* - il lavoro autonomo dello studente necessario ad integrare quanto svolto in classe.

Luiss provides e-learning content to support frontal teaching and facilitating - in a blended learning perspective - the student's independent work to integrate what has been done in the classroom.

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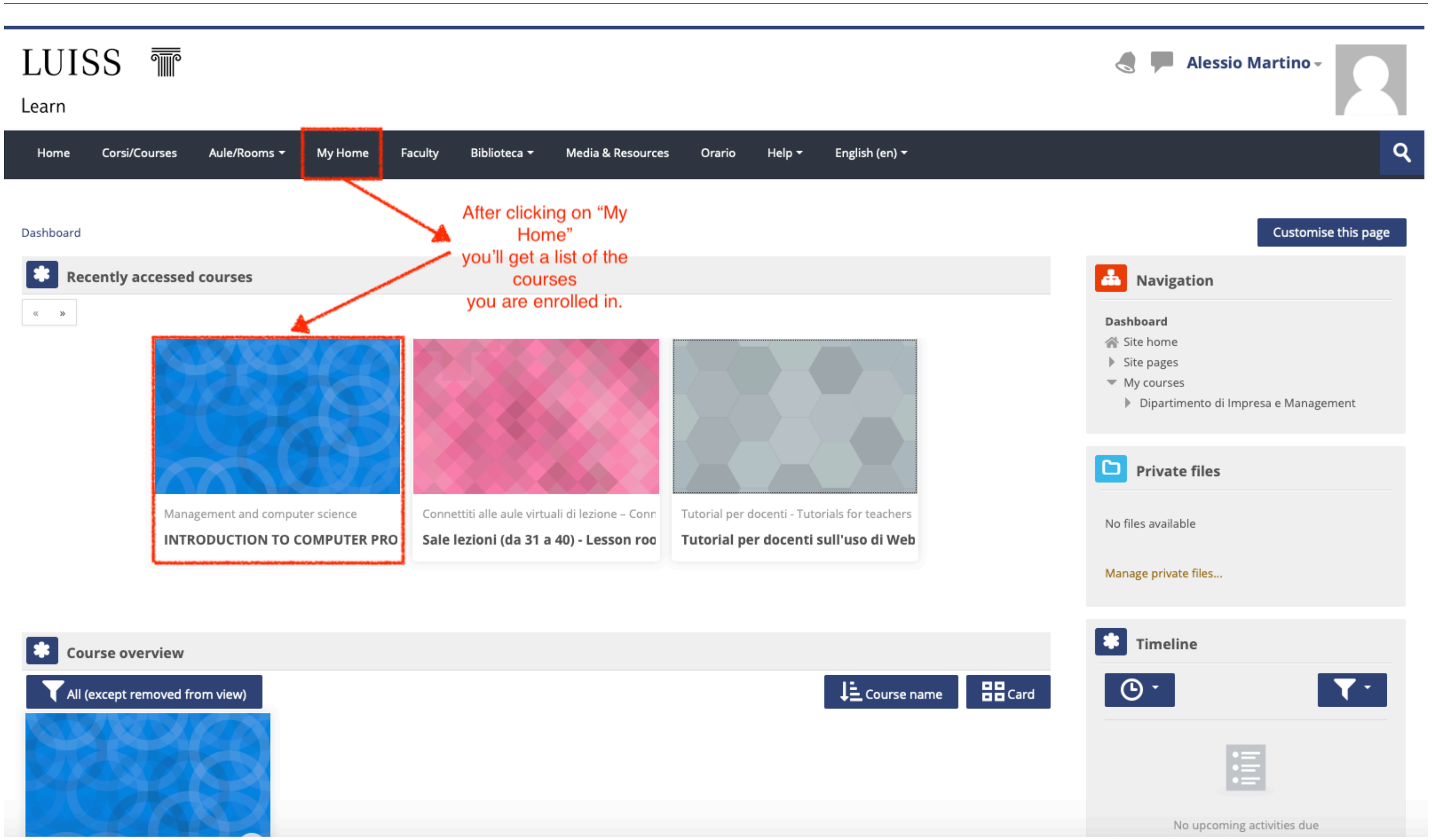
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Personnel

Teachers

- Teacher: Emilio Coppa
    - Position: Assistant Professor Tenure-Track @ LUISS
    - Research topics: software testing, vulnerability detection
    - Extra: Team Manager of TeamItaly, the italian cybersecurity team
    - Other info: https://ecoppa.github.io
    - Office Hours: send me an e-mail so we can arrange a time slot.
    - Contact: `ecoppa@luiss.it`

- Teacher: Alessio Martino
    - Position: Assistant Professor @ LUISS
    - Research topics: machine learning, big data analysis, computational biology
    - Office Hours: send me an e-mail so we can arrange a time slot.
    - Contact: `amartino@luiss.it`

Teaching assistants

- TA: Giorgio Piccardo
  - Office Hours: you can ask info in his first Lab session.
  - Contact: `gpiccardo@luiss.it`

- TA: Fabio Angeletti
  - Office Hours: you can ask info in his first Lab session.
  - Contact: `fangeletti@luiss.it`

Course Info

Course Goals

- The course is intended to teach basic programming concepts to students with no prior coding experience, developing an attitude towards computational thinking.

- It will provide the students with:

    - an understanding of the role that computation can play in solving problems
    - the ability to write small programs to accomplish useful goals

Course Roadmap

The course will familiarise the students with computer programming and will cover the following topics:

- Introduction to computation
- Introduction to programming languages
- Python basics
- Variables, data types, expressions
- Control structures
- Repetition structures
- Input / output
- Functions and modules
- Recursion
- Strings
- Lists
- Dictionaries
- Object-oriented programming and classes

Organization

Lectures + Lab Sessions: *Laptops at your fingertips, always!*

- Monday @ 14:45–16:15
  Several rooms in Viale Romania
  (often) Lab Session with TA

- Tuesday @ 10:00–11:30
  A308 (Viale Romania)
  (often) Theory lectures & some exercises

- Wednesday @ 15:45–17:15
  A211 (Viale Romania)
  (often) Lab Session with TA

Exams

Written exam:

- Coding (70% of the overall grade)
- Multiple-choice Tests (30% of the overall grade)

In light of the LUISS Continuous Assessment verification method, we will be having 2 *in itinere* exams:

- First Intermediate Test (tentative date: 23/10/2024)
- Second Intermediate Test (tentative date: 26/11/2024)

that replace the Coding test and the Multiple-choice test. Students that will not take (or pass) the intermediate tests during the course are required to take the full test at the end of the course.

Exam structure

Each intermediate test is a short exam on the topics covered so far. It will consist of a mixture of coding exercises and theory questions (multiple-choice quiz).

Rules (in brief):

- the sum of the coding parts between the two tests yields 31
- the sum of the theory parts between the two tests yields 31
- a score greater than or equal to 16 in each part is enough for successfully passing the coding + theory tests

Extra bonus points:

- pass the coding + theory test via intermediate tests, you'll get an extra bonus point
- if you pass the overall exam (coding + theory + group project) within the Winter session, you'll get an another bonus point

Computer Programming

Why computer programming?

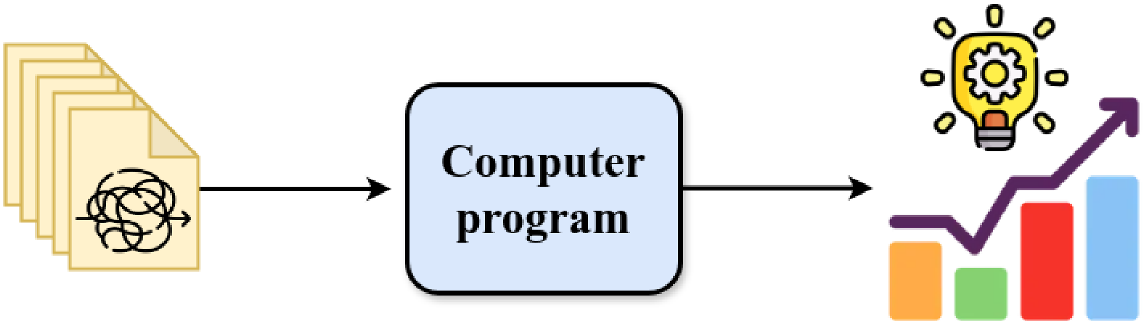
Most everyday tasks require to perform:

- repetitive...
- long sequences of...
- simple steps

This scenario is *exactly* where a computer excels!

> *Computers are incredibly fast, accurate, and stupid. Human beings are incredibly slow, inaccurate, and brilliant. Together they are powerful beyond imagination.*
>
> -- *Albert Einstein*

Why do ***you*** need to learn computer programming?



It give us the means to automatically analyze the data and extract knowledge from it.

Solving a task → Algorithm → Recipe followed by a computer

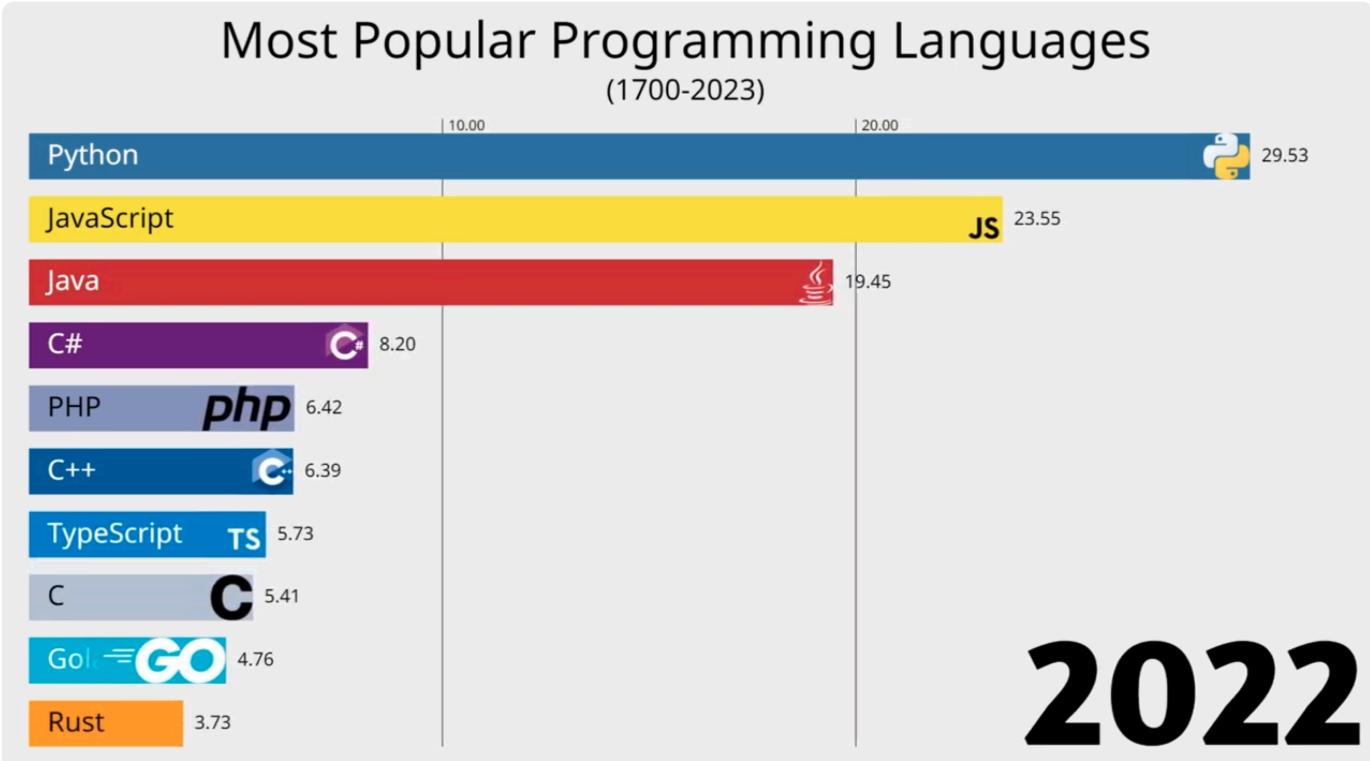
An algorithm features:

- a sequence of simple steps
- flow of control to specify when each step must be executed
- a way of determining when to stop.

...just like a recipe!

However, we cannot write an algorithm in our own natural language (e.g., english). To make an algorithm interpretable by a computer, we need to learn a programming language.

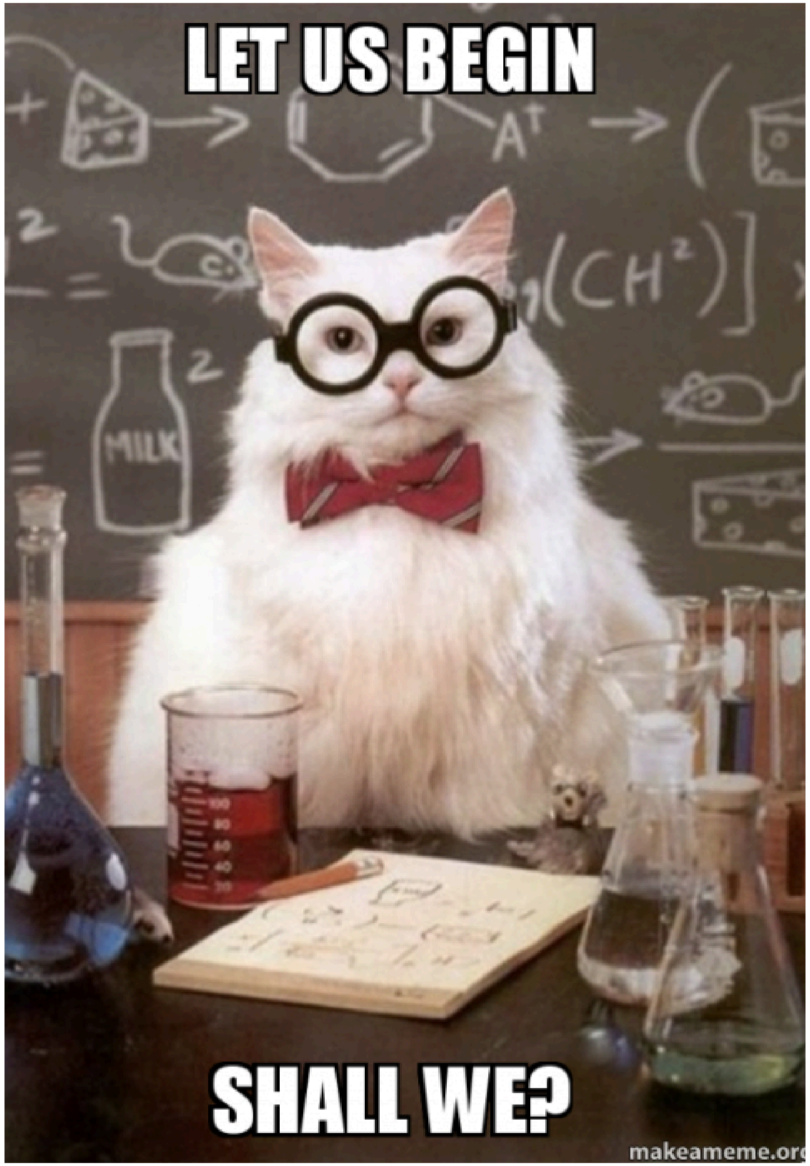
Most Popular Programming Languages



Which Computer Programming Language?

Benefits:

- Easy to learn: designed to be readable, quick to learn
- High-level language: abstracting away most low-level aspects of a computer
- Large community and great support: several libraries allow to build complex projects with just a few lines of code
- Adequate for most tasks: it shows its limits when performance is essential and low-level details play a role. You will hardly work on anything that cannot be implemented in Python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SHALL WE?
makeameme.org